

BS-CH101 / BS-CH201 — Chemistry-I

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Module 4 Long Notes: Free Energy and Chemical Equilibria

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1. First and Second Laws of Thermodynamics

Thermodynamics is the study of energy transformations. The *first law* states that energy is conserved; the *second law* states that any spontaneous process increases the total entropy of the universe. Together they define the direction and limits of all chemical and physical processes.

1.1 System, Surroundings, and Universe

Term	Definition	Types
System	The part under study	Open (matter + energy exchange), closed (energy only), isolated (neither)
Surroundings	Everything outside the system	Provides or absorbs heat and work
Universe	System + surroundings	Total energy and entropy defined for universe

1.2 Internal Energy and the First Law

First Law of Thermodynamics

$$\Delta U = q + w$$

ΔU = change in internal energy; q = heat absorbed by system (positive when absorbed); w = work done on system.

For expansion against constant pressure: $w = -P_{\text{ext}} \Delta V$.

At constant volume: $\Delta U = q_v$.

At constant pressure: $q_p = \Delta H = \Delta U + P\Delta V = \Delta U + \Delta n_g RT$ (for ideal gases).

1.3 Enthalpy and Heat Capacity

Enthalpy and Heat Capacity Relations

$$H = U + PV \Rightarrow \Delta H = \Delta U + \Delta(PV)$$

$$C_P = \left(\frac{\partial H}{\partial T} \right)_P, \quad C_V = \left(\frac{\partial U}{\partial T} \right)_V, \quad C_P - C_V = R \text{ (ideal gas)}$$

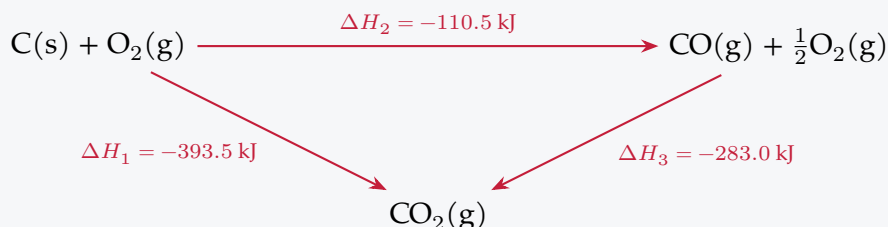
For a process at constant P : $q_P = nC_P \Delta T$.

For a process at constant V : $q_V = nC_V \Delta T$.

1.4 Hess's Law and Standard Enthalpies

Hess's law: The enthalpy change of a reaction is independent of the pathway; only the initial and final states matter. Thermochemical equations can be combined algebraically to find ΔH of reactions that are difficult to measure directly.

Hess's Law Energy Cycle for Combustion of Carbon



Solved Example: Hess's Law

Problem: Find ΔH for $\text{C(s)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO(g)}$ using:

1. $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}, \Delta H_1 = -393.5 \text{ kJ mol}^{-1}$
2. $\text{CO(g)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}, \Delta H_2 = -283.0 \text{ kJ mol}^{-1}$

Solution: Reverse equation 2 and add:

$$\Delta H = \Delta H_1 - \Delta H_2 = -393.5 - (-283.0) = -110.5 \text{ kJ mol}^{-1}$$

1.5 Entropy: The Second Law

Entropy S is a state function that measures the dispersal of energy among available microstates. The **second law** states that for any spontaneous process, $\Delta S_{\text{universe}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$. For a reversible process, $\Delta S_{\text{universe}} = 0$.

Entropy Expressions

$$dS = \frac{\delta q_{\text{rev}}}{T}, \quad \Delta S = \int_1^2 \frac{\delta q_{\text{rev}}}{T}$$

For a reversible isothermal expansion of ideal gas: $\Delta S = nR \ln \frac{V_2}{V_1}$.

For heating at constant P : $\Delta S = nC_P \ln \frac{T_2}{T_1}$.

For heating at constant V : $\Delta S = nC_V \ln \frac{T_2}{T_1}$.

Clausius inequality: $\oint \frac{\delta q}{T} \leq 0$ (equality for reversible cycle).

1.6 Third Law and Absolute Entropy

Third law of thermodynamics: The entropy of a perfect crystalline substance is zero at absolute zero ($S = 0$ at $T = 0$ K). This provides an absolute reference scale for entropy.

Standard	molar	entropy	S°	at	298	K	(J	mol ⁻¹	K ⁻¹):
Substance	S°			Substance	S°				
C(graphite)	5.7			H ₂ O(l)	69.9				
H ₂ (g)	130.7			CO ₂ (g)	213.8				
O ₂ (g)	205.2			NaCl(s)	72.1				

Standard entropy change of reaction: $\Delta S^\circ_{\text{rxn}} = \sum n_p S^\circ(\text{products}) - \sum n_r S^\circ(\text{reactants})$.

1.7 Work in Reversible and Irreversible Expansion

This is a 5-mark derivation in nearly every MAKAUT paper.

Work Done by an Ideal Gas

Reversible isothermal expansion (maximum work):

$$w_{\text{rev}} = -nRT \ln \frac{V_2}{V_1} = -nRT \ln \frac{P_1}{P_2}$$

Irreversible expansion against constant external pressure:

$$w_{\text{irrev}} = -P_{\text{ext}} \Delta V = -P_{\text{ext}} (V_2 - V_1)$$

Free expansion (into vacuum, $P_{\text{ext}} = 0$): $w = 0$.

Rule: $|w_{\text{rev}}| > |w_{\text{irrev}}|$ always — reversible processes deliver maximum work.

Solved Example: Reversible vs Irreversible Work

1 mol ideal gas at 300 K expands from $V_1 = 10$ L to $V_2 = 30$ L.

Reversible:

$$w_{\text{rev}} = -nRT \ln \frac{V_2}{V_1} = -(1)(8.314)(300) \ln 3 = -2494 \times 1.099 = -2741 \text{ J}$$

Irreversible against $P_{\text{ext}} = P_2 = nRT/V_2 = (1)(8.314)(300)/30 \times 10^{-3} = 83140$ Pa:

$$w_{\text{irrev}} = -(83140)(30 - 10) \times 10^{-3} = -1663 \text{ J}$$

Reversible work (2741 J) is greater than irreversible work (1663 J) — confirms second law.

1.8 Entropy of Phase Transition and Trouton's Rule

At a phase transition (melting, boiling), temperature is constant and heat flows reversibly:

$$\Delta S_{\text{vap}} = \frac{\Delta H_{\text{vap}}}{T_b}, \quad \Delta S_{\text{fus}} = \frac{\Delta H_{\text{fus}}}{T_m}$$

Trouton's rule: For most non-associated liquids, $\Delta S_{\text{vap}} \approx 85 \text{ J mol}^{-1} \text{ K}^{-1}$ at the normal boiling point. Water deviates ($\Delta S_{\text{vap}} \approx 109 \text{ J mol}^{-1} \text{ K}^{-1}$) because its extensive H-bonding creates extra order in the liquid phase.

1.9 Kirchhoff's Equation: ΔH Variation with Temperature

Enthalpy of a reaction changes with temperature because reactant and product heat capacities differ.

Kirchhoff's Equation

$$\frac{d(\Delta H)}{dT} = \Delta C_P = \sum n_p C_P(\text{products}) - \sum n_r C_P(\text{reactants})$$

Integrated form (assuming ΔC_P is constant):

$$\Delta H_{T_2} = \Delta H_{T_1} + \Delta C_P (T_2 - T_1)$$

Exam Points (Section 1)

- First law: $\Delta U = q + w$; at const P : $\Delta H = q_P$.
- $C_P - C_V = R$ for ideal gas; $\Delta H = \Delta U + \Delta n_g RT$ for gas reactions.
- Hess's law: ΔH is path-independent; combine equations algebraically.
- Second law: $\Delta S_{\text{univ}} > 0$ for spontaneous; $= 0$ reversible.
- Third law: $S = 0$ at 0 K for perfect crystal — provides absolute entropy scale.

2. Gibbs Free Energy and Spontaneity

Gibbs free energy $G = H - TS$ is the thermodynamic potential that determines spontaneity at constant temperature and pressure. It combines both the enthalpic driving force (energy minimisation) and the entropic driving force (disorder maximisation) into a single criterion.

2.1 Derivation of the Gibbs Free Energy Criterion

At constant T and P , the second law requires $\Delta S_{\text{univ}} \geq 0$. Since $\Delta S_{\text{surr}} = -\Delta H/T$ (heat lost by system is gained by surroundings reversibly):

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} - \frac{\Delta H}{T} \geq 0 \quad \Rightarrow \quad T\Delta S - \Delta H \geq 0 \quad \Rightarrow \quad \Delta H - T\Delta S \leq 0$$

Defining $\Delta G = \Delta H - T\Delta S$:

Gibbs Free Energy Criterion

$$\Delta G = \Delta H - T\Delta S$$

- $\Delta G < 0$: spontaneous (product-favoured).
 - $\Delta G = 0$: system at equilibrium.
 - $\Delta G > 0$: non-spontaneous (reactant-favoured under those conditions).
- ΔG equals the maximum non-expansion (useful) work: $w_{\max} = \Delta G$.

2.2 The Four Spontaneity Cases

The sign of $\Delta G = \Delta H - T\Delta S$ depends on temperature through the entropy term.

ΔH	ΔS	Sign of ΔG	Spontaneity
–	+	Always –	Spontaneous at all temperatures
+	–	Always +	Non-spontaneous at all temperatures
–	–	– at low T , + at high T	Spontaneous at low temperature
+	+	+ at low T , – at high T	Spontaneous at high temperature

Crossover temperature (where $\Delta G = 0$):

$$T^* = \frac{\Delta H}{\Delta S}$$

Solved Example: Crossover Temperature

For the decomposition of $\text{CaCO}_3(s) \rightarrow \text{CaO}(s) + \text{CO}_2(g)$: $\Delta H^\circ = +178 \text{ kJ mol}^{-1}$, $\Delta S^\circ = +165 \text{ J mol}^{-1} \text{ K}^{-1} = +0.165 \text{ kJ mol}^{-1} \text{ K}^{-1}$.

$$T^* = \frac{178}{0.165} = 1079 \text{ K}$$

Below 1079 K the reaction is non-spontaneous ($\Delta G > 0$); above it becomes spontaneous ($\Delta G < 0$). This explains why limestone kilns operate above $\sim 800^\circ\text{C}$.

2.3 Standard Free Energy of Formation

ΔG_f° is the Gibbs free energy change when one mole of a compound is formed from its elements in their standard states at 298 K and 1 bar. By definition, $\Delta G_f^\circ = 0$ for all elements in their standard states.

$$\Delta G_{\text{rxn}}^\circ = \sum n_p \Delta G_f^\circ(\text{products}) - \sum n_r \Delta G_f^\circ(\text{reactants})$$

Substance	ΔG_f° (kJ/mol)	Substance	ΔG_f° (kJ/mol)
H ₂ O(l)	-237.1	CO ₂ (g)	-394.4
CO(g)	-137.2	NH ₃ (g)	-16.5
HCl(g)	-95.3	NO(g)	+86.6

2.4 Temperature Dependence: Gibbs–Helmholtz Equation

How does ΔG change with temperature?

Gibbs–Helmholtz Equation

$$\left(\frac{\partial(\Delta G/T)}{\partial T} \right)_P = -\frac{\Delta H}{T^2}$$

Integrated form (assuming ΔH is independent of T):

$$\frac{\Delta G_2}{T_2} - \frac{\Delta G_1}{T_1} = -\Delta H \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

Used to calculate ΔG at a new temperature T_2 from a known value at T_1 .

Solved Example: ΔG at New Temperature

$\Delta G_{298}^\circ = -33.0 \text{ kJ mol}^{-1}$, $\Delta H^\circ = -92.0 \text{ kJ mol}^{-1}$ for $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$. Find ΔG° at 500 K:

$$\frac{\Delta G_{500}^\circ}{500} - \frac{-33000}{298} = -(-92000) \left(\frac{1}{500} - \frac{1}{298} \right)$$

$$\frac{\Delta G_{500}^\circ}{500} + 110.7 = 92000 \times \frac{202}{149000} = 124.8$$

$$\Delta G_{500}^\circ = (124.8 - 110.7) \times 500 = +7050 \text{ J mol}^{-1} = +7.1 \text{ kJ mol}^{-1}$$

The reaction becomes non-spontaneous above $\sim 460 \text{ K}$ (industrial Haber process must use a catalyst to achieve acceptable rate below this temperature).

Exam Points (Section 2)

- Derive $\Delta G = \Delta H - T\Delta S$ from the second law at constant T , P .
- Quote all 4 sign-of- ΔG cases; always give the temperature condition.
- Crossover temperature: $T^* = \Delta H/\Delta S$ (units: J cancel, gives K directly if both in J).
- $\Delta G_{\text{rxn}}^\circ$ from formation free energies: products minus reactants.
- Gibbs–Helmholtz: $(\partial(\Delta G/T)/\partial T)_P = -\Delta H/T^2$.

3. Free Energy and Chemical Equilibrium

At equilibrium, the Gibbs free energy of a reacting mixture reaches its minimum value. The standard free energy change ΔG° is directly related to the equilibrium constant K through $\Delta G^\circ = -RT \ln K$, providing the fundamental link between thermodynamics and equilibrium chemistry.

3.1 Chemical Potential and Reaction Quotient

For an ideal gas species i at partial pressure P_i , the chemical potential is:

$$\mu_i = \mu_i^\circ + RT \ln \frac{P_i}{P^\circ}$$

For a reaction $aA + bB \rightarrow cC + dD$, the free energy change is:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

where the reaction quotient $Q = \frac{[C]^c[D]^d}{[A]^a[B]^b}$ uses current concentrations (or partial pressures).

3.2 Relationships Between K_p , K_c , K_x

Three forms of the equilibrium constant are commonly used. They are interrelated through the ideal gas law.

Equilibrium Constant Relationships

$$K_p = K_c (RT)^{\Delta n_g}, \quad K_p = K_x P^{\Delta n_g}$$

K_p = equilibrium constant in terms of partial pressures (bar or atm).

K_c = equilibrium constant in terms of molar concentrations (mol L^{-1}).

K_x = equilibrium constant in terms of mole fractions (dimensionless).

Δn_g = moles of gaseous products – moles of gaseous reactants.

When $\Delta n_g = 0$: $K_p = K_c = K_x$ (e.g. $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$).

Solved Example: K_p from K_c

For $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ at 500 K, $K_c = 6.0 \times 10^{-2} \text{ L}^2 \text{ mol}^{-2}$.

$\Delta n_g = 2 - (1 + 3) = -2$.

$$K_p = K_c (RT)^{\Delta n_g} = 6.0 \times 10^{-2} \times (0.0821 \times 500)^{-2} = 6.0 \times 10^{-2} \times (41.05)^{-2} = 3.55 \times 10^{-5} \text{ atm}^{-2}$$

$K_p < K_c$ because $\Delta n_g < 0$ (fewer moles of gas at equilibrium).

3.3 Derivation of $\Delta G^\circ = -RT \ln K$

Free Energy and Equilibrium Constant

$$\Delta G = \Delta G^\circ + RT \ln Q$$

At equilibrium: $\Delta G = 0$ and $Q = K$, so:

$$0 = \Delta G^\circ + RT \ln K \quad \Rightarrow \quad \boxed{\Delta G^\circ = -RT \ln K = -2.303 RT \log K}$$

At 298 K: $RT = 2.478 \text{ kJ mol}^{-1}$, so a 10-fold change in K changes ΔG° by 5.71 kJ mol^{-1} .

Relationship between ΔG° , K , and reaction direction:

ΔG°	K	Meaning
$\ll 0$ (e.g. -100 kJ)	$K \gg 1$	Products strongly favoured
≈ 0	$K \approx 1$	Reactants and products in comparable amounts
$\gg 0$ (e.g. $+100 \text{ kJ}$)	$K \ll 1$	Reactants strongly favoured

Solved Example: K from ΔG°

For $\text{N}_2\text{O}_4(g) \rightleftharpoons 2\text{NO}_2(g)$ at 298 K, $\Delta G^\circ = +4.8 \text{ kJ mol}^{-1}$.

$$\ln K = \frac{-\Delta G^\circ}{RT} = \frac{-4800}{8.314 \times 298} = -1.937$$

$$K = e^{-1.937} = 0.144$$

$K < 1$: at equilibrium, N_2O_4 predominates. At higher temperatures K rises and NO_2 increases.

3.4 Van't Hoff Equation: Temperature Dependence of K

Combining $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ with $\Delta G^\circ = -RT \ln K$:

$$-RT \ln K = \Delta H^\circ - T\Delta S^\circ \quad \Rightarrow \quad \ln K = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

Differentiating with respect to T :

Van't Hoff Equation

$$\frac{d \ln K}{dT} = \frac{\Delta H^\circ}{RT^2}$$

Integrated form:

$$\ln \frac{K_2}{K_1} = -\frac{\Delta H^\circ}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

Exothermic reaction ($\Delta H^\circ < 0$): K decreases as T increases.

Endothermic reaction ($\Delta H^\circ > 0$): K increases as T increases.

Solved Example: Van't Hoff Equation

For the Haber process, $K_1 = 977$ at 300 K and $\Delta H^\circ = -92 \text{ kJ mol}^{-1}$. Find K_2 at 500 K.

$$\ln \frac{K_2}{977} = -\frac{-92000}{8.314} \left(\frac{1}{500} - \frac{1}{300} \right) = 11062 \times (-1.333 \times 10^{-3}) = -14.75$$

$$K_2 = 977 \times e^{-14.75} = 977 \times 3.93 \times 10^{-7} = 3.84 \times 10^{-4}$$

The equilibrium constant drops by a factor of $\sim 2.5 \times 10^6$ from 300 K to 500 K — explaining why the Haber process is thermodynamically challenged at high temperature.

3.5 Le Chatelier's Principle

Le Chatelier's principle: When a system at equilibrium is disturbed, it shifts in the direction that partially counteracts the disturbance and re-establishes equilibrium.

Perturbation	Direction of shift	Effect on K
Add reactant	Forward (towards products)	K unchanged
Remove product	Forward	K unchanged
Increase pressure (gas reaction)	Towards side with fewer moles of gas	K unchanged
Increase temperature	Towards endothermic side	K changes (van't Hoff)
Add inert gas (constant V)	No shift	K unchanged
Add catalyst	Equilibrium reached faster; no shift	K unchanged

Exam Points (Section 3)

- Derive $\Delta G^\circ = -RT \ln K$ from $\Delta G = \Delta G^\circ + RT \ln Q$ at equilibrium.
- If $\Delta G^\circ < 0$: $K > 1$ (products favoured). If $\Delta G^\circ > 0$: $K < 1$.
- Van't Hoff: $\ln(K_2/K_1) = -(\Delta H^\circ/R)(1/T_2 - 1/T_1)$.
- Exothermic reaction: K decreases with temperature; endothermic: K increases.
- Le Chatelier: temperature change alters K ; other disturbances shift position but not K .

4. Electrochemistry: Cell Potentials and the Nernst Equation

Electrochemistry studies the interconversion of chemical and electrical energy. In a **galvanic cell**, a spontaneous redox reaction drives an electric current. The Gibbs free energy of the cell reaction is directly converted to electrical work: $\Delta G = -nFE$.

4.1 Galvanic vs Electrolytic Cells

Feature	Galvanic cell	Electrolytic cell
Energy source	Spontaneous chemical reaction	External electrical supply
ΔG	Negative (spontaneous)	Positive (non-spontaneous)
Anode polarity	Negative (oxidation)	Positive (oxidation)
Cathode polarity	Positive (reduction)	Negative (reduction)
Applications	Batteries, fuel cells	Electroplating, electrolysis

4.2 Standard Hydrogen Electrode and Standard Potentials

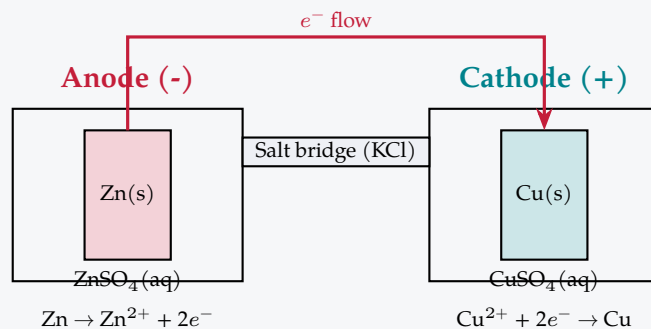
The **Standard Hydrogen Electrode (SHE)** consists of Pt immersed in 1 M H^+ with H_2 gas at 1 atm. Its electrode potential is defined as exactly 0.000 V at all temperatures. All standard electrode potentials E° are measured relative to SHE.

Half-reaction (reduction)	E° (V)
$\text{F}_2 + 2\text{e}^- \rightarrow 2\text{F}^-$	+2.87
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	+1.51
$\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$	+1.36
$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	+0.34
$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	0.00 (ref)
$\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$	-0.44
$\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$	-0.76
$\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$	-2.71

Cell EMF: $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$.

The species with higher E° acts as cathode (is reduced); lower E° acts as anode (is oxidised).

Galvanic Cell: Daniell Cell (Zn–Cu)

4.3 Relationship Between ΔG and Cell EMF

Free Energy and EMF

$$\Delta G = -nFE, \quad \Delta G^\circ = -nFE^\circ$$

n = number of electrons transferred; $F = 96485 \text{ C mol}^{-1}$ (Faraday constant).

Combining with $\Delta G^\circ = -RT \ln K$:

$$E^\circ = \frac{RT}{nF} \ln K = \frac{0.0257}{n} \ln K = \frac{0.0591}{n} \log K \quad (298 \text{ K})$$

4.4 Derivation of the Nernst Equation

Starting from $\Delta G = \Delta G^\circ + RT \ln Q$ and substituting $\Delta G = -nFE$ and $\Delta G^\circ = -nFE^\circ$:

$$-nFE = -nFE^\circ + RT \ln Q$$

Dividing by $-nF$:

Nernst Equation

$$E = E^\circ - \frac{RT}{nF} \ln Q = E^\circ - \frac{0.0591}{n} \log Q \quad (298 \text{ K})$$

At equilibrium, $E = 0$ and $Q = K$, confirming $E^\circ = (0.0591/n) \log K$.

Solved Example: Daniell Cell EMF via Nernst

Zn/Cu cell: $E^\circ_{\text{cell}} = 0.34 - (-0.76) = 1.10 \text{ V}$, $n = 2$. If $[\text{Zn}^{2+}] = 0.10 \text{ M}$ and $[\text{Cu}^{2+}] = 2.0 \text{ M}$:

$$Q = \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} = \frac{0.10}{2.0} = 0.050$$

$$E = 1.10 - \frac{0.0591}{2} \log(0.050) = 1.10 - 0.02955 \times (-1.301) = 1.10 + 0.038 = 1.138 \text{ V}$$

Lower $[\text{Zn}^{2+}]$ and higher $[\text{Cu}^{2+}]$ increases the driving force above the standard value.

Solved Example: Equilibrium Constant from E°

For the Daniell cell ($E^\circ = 1.10\text{ V}$, $n = 2$):

$$\log K = \frac{n E^\circ}{0.0591} = \frac{2 \times 1.10}{0.0591} = 37.23$$

$$K = 10^{37.23} \approx 1.7 \times 10^{37}$$

An enormous K confirms the reaction is essentially complete under standard conditions.

4.5 Concentration Cell

A concentration cell uses the same electrode material in solutions of different concentrations. The EMF arises purely from the concentration difference.

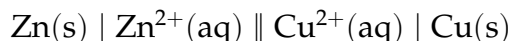
Concentration Cell EMF

$$E = \frac{0.0591}{n} \log \frac{C_{\text{high}}}{C_{\text{low}}}$$

At equal concentrations $E = 0$; the cell drives ions from high to low concentration.

4.6 Cell Notation and Standard Cell Notation

Cells are written with the anode on the left and cathode on the right, separated by a double vertical line (salt bridge):

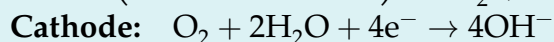
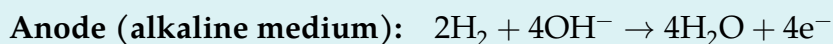


Single vertical line $|$ = phase boundary; double $\parallel$ = salt bridge.

$$E^\circ_{\text{cell}} = E^\circ(\text{right half-cell}) - E^\circ(\text{left half-cell}) = 0.34 - (-0.76) = 1.10\text{ V}.$$

4.7 Fuel Cells

A fuel cell is a galvanic device that converts the chemical energy of a fuel (usually H_2) directly to electricity without combustion.

Hydrogen–Oxygen Fuel Cell Reactions**Advantages of fuel cells over heat engines:**

- Efficiency not limited by Carnot efficiency ($\eta_{\text{max}} = 1 - T_c/T_h$); can exceed 60%.
- No moving parts in the electrochemical stack.
- Zero local emissions (H_2 - O_2 cell produces only water).
- Used in spacecraft, buses, and stationary power plants.

Exam Points (Section 4)

- $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$; higher E° is cathode.
- $\Delta G = -nFE$; $\Delta G < 0$ when $E > 0$ (spontaneous cell).
- Derive Nernst from $\Delta G = \Delta G^\circ + RT \ln Q$.
- $\log K = nE^\circ/0.0591$ at 298 K.
- Concentration cell: EMF from $\log(C_{\text{high}}/C_{\text{low}})$; same electrodes.

5. Acid–Base, Redox, and Solubility Equilibria

Three categories of aqueous equilibria are central to engineering chemistry: (i) acid–base equilibria (proton transfer), (ii) redox equilibria (electron transfer), and (iii) solubility equilibria (precipitation). All three are governed by $\Delta G^\circ = -RT \ln K$.

5.1 Brønsted–Lowry Acid–Base Theory

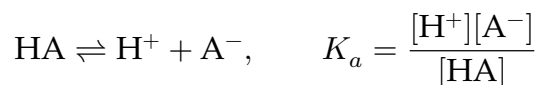
A **Brønsted–Lowry acid** donates a proton; a **base** accepts one. Every acid has a conjugate base (formed after proton loss) and every base has a conjugate acid. This theory covers aqueous and non-aqueous systems, unlike Arrhenius theory.

Water autoionisation: $\text{H}_2\text{O} + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{OH}^-$; $K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 1.0 \times 10^{-14}$ at 298 K.

$\text{pH} + \text{pOH} = 14$ at 298 K.

5.2 Weak Acid Equilibrium and pH Derivation

For a weak acid HA in water:



If initial concentration is C and degree of dissociation $\alpha \ll 1$:

$$[\text{H}^+] = \sqrt{K_a C}, \quad \text{pH} = \frac{1}{2}(\text{p}K_a - \log C)$$

Solved Example: pH of Acetic Acid

0.10 M acetic acid, $K_a = 1.8 \times 10^{-5}$:

$$[\text{H}^+] = \sqrt{1.8 \times 10^{-5} \times 0.10} = \sqrt{1.8 \times 10^{-6}} = 1.34 \times 10^{-3} \text{ M}$$

$$\text{pH} = -\log(1.34 \times 10^{-3}) = 2.87$$

Verify with formula: $\text{pH} = \frac{1}{2}(4.74 - (-1)) = 2.87$. ✓

5.3 Henderson–Hasselbalch Equation and Buffers

A buffer solution resists pH change. It consists of a weak acid and its conjugate base (or weak base and conjugate acid).

Henderson–Hasselbalch Equation

$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

Derivation: Take $-\log$ of $K_a = [\text{H}^+][\text{A}^-]/[\text{HA}]$: $\text{p}K_a = \text{pH} - \log([\text{A}^-]/[\text{HA}])$; rearrange. Buffer capacity is maximum when $[\text{A}^-] = [\text{HA}]$, i.e. $\text{pH} = \text{p}K_a$.

Solved Example: Buffer pH and Its Change on Adding Acid

Acetate buffer: $[\text{CH}_3\text{COOH}] = 0.20 \text{ M}$, $[\text{CH}_3\text{COONa}] = 0.30 \text{ M}$, $\text{p}K_a = 4.74$:

$$\text{pH} = 4.74 + \log \frac{0.30}{0.20} = 4.74 + 0.18 = 4.92$$

Adding 0.01 mol HCl to 1 L: acid consumes 0.01 mol of CH_3COO^- :

$$[\text{A}^-] = 0.29 \text{ M}, \quad [\text{HA}] = 0.21 \text{ M}, \quad \text{pH} = 4.74 + \log \frac{0.29}{0.21} = 4.74 + 0.14 = 4.88$$

pH changed by only 0.04 despite adding strong acid — the buffer works.

5.4 Biological Importance of Buffers

- Blood pH (7.35–7.45) maintained by the bicarbonate buffer: $\text{H}_2\text{CO}_3/\text{HCO}_3^-$ ($\text{p}K_a = 6.35$) reinforced by respiratory control of CO_2 .
- Intracellular pH maintained by phosphate buffer ($\text{H}_2\text{PO}_4^-/\text{HPO}_4^{2-}$, $\text{p}K_a = 7.2$).
- Industrial buffers: electroplating baths, pharmaceutical formulations.

5.5 Lewis Acid–Base Theory

Lewis acid: Electron-pair acceptor. **Lewis base:** Electron-pair donor. This is the broadest acid–base definition and covers reactions without proton transfer, such as complex-ion formation.

Concept	Arrhenius	Brønsted–Lowry and Lewis
Acid	Produces H^+ in water	H^+ donor / electron-pair acceptor
Base	Produces OH^- in water	H^+ acceptor / electron-pair donor
Scope	Aqueous only	Aqueous and non-aqueous
Example	HCl, NaOH	NH_3 : Lewis base; BF_3 : Lewis acid

Lewis examples: $\text{BF}_3 + \text{NH}_3 \rightarrow \text{F}_3\text{B}:\text{NH}_3$ (adduct); metal-ligand complex formation $\text{Cu}^{2+} + 4\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4]^{2+}$ (Cu^{2+} = Lewis acid, NH_3 = Lewis base).

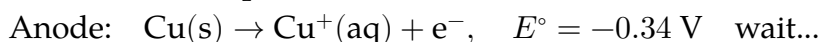
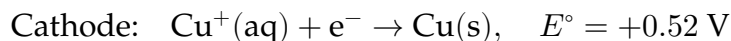
5.6 Redox Equilibria and Disproportionation

Disproportionation: A reaction in which the same element is simultaneously oxidised and reduced. Example: $2\text{Cu}^+(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + \text{Cu}(\text{s})$.

Predicting feasibility of redox reactions using E° :

- $E^\circ_{\text{cell}} > 0$ ($\Delta G^\circ < 0$): reaction spontaneous.
- $E^\circ_{\text{cell}} < 0$: non-spontaneous in the written direction.

Disproportionation of Cu^+ :



Actually using the two half-reactions: $\text{Cu}^+ + \text{e}^- \rightarrow \text{Cu}$ (+0.52 V) and $\text{Cu}^{2+} + \text{e}^- \rightarrow \text{Cu}^+$ (+0.16 V). $E^\circ_{\text{cell}} = 0.52 - 0.16 = +0.36 \text{ V} > 0$, so Cu^+ is unstable to disproportionation in aqueous solution.

5.7 Solubility Product K_{sp}

For a sparingly soluble salt $\text{M}_m\text{X}_n(\text{s}) \rightleftharpoons m\text{M}^{n+} + n\text{X}^{m-}$:

Solubility Product

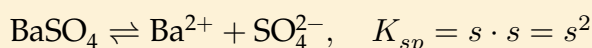
$$K_{sp} = [\text{M}^{n+}]^m [\text{X}^{m-}]^n$$

Ionic product (IP): $[\text{M}^{n+}]^m [\text{X}^{m-}]^n$ using current concentrations.

- $\text{IP} < K_{sp}$: unsaturated, no precipitation.
- $\text{IP} = K_{sp}$: saturated, equilibrium.
- $\text{IP} > K_{sp}$: supersaturated, precipitation occurs.

Solved Example: Solubility from K_{sp}

$K_{sp}(\text{BaSO}_4) = 1.1 \times 10^{-10}$ at 298 K. Find molar solubility s in pure water.



$$s = \sqrt{1.1 \times 10^{-10}} = 1.05 \times 10^{-5} \text{ mol L}^{-1}$$

In 0.10 M Na_2SO_4 (common ion effect):

$$K_{sp} = s \cdot (0.10 + s) \approx 0.10 s \quad \Rightarrow \quad s = \frac{1.1 \times 10^{-10}}{0.10} = 1.1 \times 10^{-9} \text{ mol L}^{-1}$$

Common ion reduces solubility by a factor of $\sim 10^4$.

Exam Points (Section 5)

- Brønsted–Lowry: acid = proton donor; base = proton acceptor.
- $K_a \times K_b = K_w = 10^{-14}$; $\text{pH} + \text{pOH} = 14$ at 298 K.
- Weak acid pH: $\text{pH} = \frac{1}{2}(\text{pK}_a - \log C)$.
- Henderson–Hasselbalch: $\text{pH} = \text{pK}_a + \log([A^-]/[HA])$; buffer works near pK_a .
- K_{sp} solubility product: $\text{IP} > K_{sp}$ gives precipitation; common ion reduces solubility.

6. Water Chemistry and Corrosion

Water chemistry deals with the dissolved ions, gases, and organic matter in natural and industrial water, especially as they affect engineering systems. **Corrosion** is the electrochemical deterioration of metals driven by a spontaneous redox reaction ($\Delta G < 0$). Both topics are directly governed by the free-energy and equilibrium principles of this module.

6.1 Hardness of Water

Hard water contains dissolved Ca^{2+} and Mg^{2+} ions, primarily as bicarbonates, chlorides, and sulphates. Hardness is expressed in parts per million (ppm) as equivalent CaCO_3 .

Hardness in ppm as CaCO_3 Equivalent

$$\text{Hardness (ppm)} = \frac{\text{mass of salt (g)}}{\text{volume of water (L)}} \times \frac{M(\text{CaCO}_3)}{M(\text{salt})} \times 1000$$

$$\text{Or: ppm} = \frac{w_{\text{CaCO}_3\text{-equiv}}}{V_{\text{water}}} \times 10^6.$$

Soft water: < 75 ppm. Moderately hard: 75–150 ppm. Hard: > 150 ppm.

Type	Cause	Removal
Temporary hardness	$\text{Ca}(\text{HCO}_3)_2$, $\text{Mg}(\text{HCO}_3)_2$	Boiling, lime addition
Permanent hardness	CaCl_2 , MgCl_2 , CaSO_4 , MgSO_4	Lime-soda, zeolite, ion exchange

6.2 Boiler Troubles

Hard water in steam boilers causes multiple engineering problems.

Problem	Cause	Effect	Remedy
Scale formation	Deposition of CaCO_3 , Mg(OH)_2 , CaSO_4	Reduced heat transfer; tube overheating and bursting	Softening; acid descaling
Sludge formation	Loose precipitates of MgCO_3 , MgCl_2	Chokes pipes; reduces efficiency	Blow-down of boiler
Priming	High dissolved solids, oil, or organic impurities	Wet steam carries water droplets; turbine erosion	Filtration; demineralisation
Foaming	Organic impurities or oil stabilise steam bubbles	Irregular operation; carryover	Anti-foam chemicals
Caustic embrittlement	Concentrated NaOH in stressed joints/cracks	Metal grain-boundary cracking and failure	Neutralise with sodium sulphate

6.3 Water Softening Methods

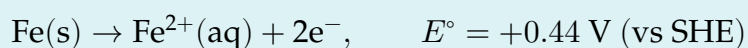
Method	Principle	Notes
Lime-soda process	Add Ca(OH)_2 and Na_2CO_3 ; precipitate CaCO_3 and Mg(OH)_2	Cheap; used for large volumes; produces sludge
Zeolite / ion exchange	Ca^{2+} and Mg^{2+} exchanged for Na^+ on zeolite resin	Portable; regenerated with NaCl; does not reduce TDS
Demineralisation	Cation exchanger (H^+ form) + anion exchanger (OH^- form)	Removes all ions; produces ultra-pure water
Reverse osmosis	Applied pressure forces water through semi-permeable membrane	Removes ions, bacteria, organics

6.4 Electrochemical Theory of Corrosion

Corrosion of iron in neutral or acidic water is an electrochemical process requiring an anode (oxidation), a cathode (reduction), an electrolyte, and a metallic conductor.

Iron Corrosion Half-Reactions

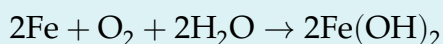
Anodic reaction (oxidation):



Cathodic reaction (oxygen reduction, neutral/alkaline):

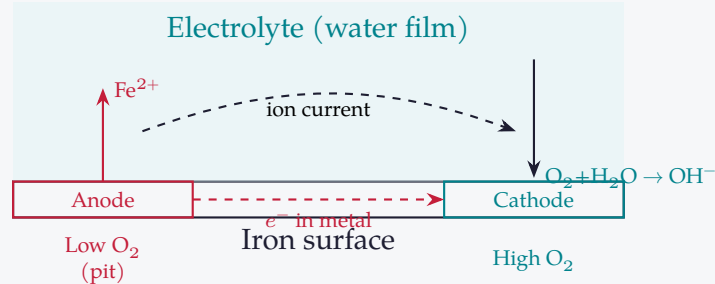


Overall:



Further air oxidation: $4\text{Fe(OH)}_2 + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{Fe(OH)}_3 \rightarrow \text{rust (hydrated Fe}_2\text{O}_3\text{)}.$

Differential Aeration Corrosion Cell



Factors accelerating corrosion: Higher temperature, lower pH, higher dissolved O_2 , electrolyte concentration, stressed metal, galvanic coupling with nobler metal.

6.5 Dissolved Oxygen, BOD, and COD

These three parameters are standard measures of water quality and appear in MAKAUT water-chemistry questions.

Dissolved Oxygen (DO): Oxygen dissolved in water; typical freshwater has $8\text{--}14\text{ mg L}^{-1}$. Falls below 5 mg L^{-1} in polluted water, harming aquatic life.

BOD (Biochemical Oxygen Demand): Oxygen consumed by microorganisms to decompose organic matter in 5 days at 20°C (BOD_5). High BOD = heavy organic pollution.

COD (Chemical Oxygen Demand): Oxygen equivalent needed to oxidise all organic and inorganic matter chemically (with $\text{K}_2\text{Cr}_2\text{O}_7$ as oxidant). $\text{COD} \geq \text{BOD}$ always.

Parameter	Clean water	Polluted water
DO (mg/L)	8–14	<5 (danger zone)
BOD (mg/L)	<1–2	>100 (sewage ~ 300)
COD (mg/L)	<10	>150

6.6 Passivation

Passivation: Formation of a thin, dense, adherent oxide film on a metal surface that blocks further corrosion. The film is self-healing when scratched in air. Examples: Al forms Al_2O_3 (3 nm thick), stainless steel forms Cr_2O_3 , Ti forms TiO_2 .

Passivation requires the oxide to be: (i) insoluble in the corrosive medium, (ii) adherent (not flaking), (iii) dense enough to prevent ion/electron diffusion.

Concentration of HNO_3 and passivation of iron: Dilute HNO_3 dissolves iron (active dissolution); concentrated HNO_3 passivates iron by forming a dense $\text{Fe}_3\text{O}_4/\text{Fe}_2\text{O}_3$ surface layer — a standard MAKAUT viva question.

6.7 Corrosion Prevention

Method	Principle	Example
Barrier coatings	Physically isolate metal from electrolyte	Paint, polymer coating, oxide passivation
Galvanisation	Sacrificial zinc oxidises preferentially to iron; even at scratches Zn protects Fe	Zn-coated steel sheets
Sacrificial anode	More active metal (Mg, Zn, Al) connected as anode; made to corrode instead of structure	Ship hulls, buried pipelines
Impressed current	External DC supply makes structure cathodic; no net oxidation	Underground pipelines, bridges
Alloying	Cr_2O_3 passive film on stainless steel blocks $\text{O}_2/\text{H}_2\text{O}$ access	Stainless steel ($> 11\% \text{ Cr}$)
Inhibitors	Adsorb on metal surface, reduce anodic/cathodic reaction rate	Boiler treatment chemicals

Sacrificial anode vs impressed current:

Feature	Sacrificial anode	Impressed current
Power source	None (galvanic cell)	External DC rectifier
Anode material	Mg, Zn, or Al alloy	Inert (platinised Ti, graphite)
Maintenance	Replace anode periodically	Monitor current; less material cost
Suitable for	Small isolated structures	Large structures, long pipelines

Exam Points (Section 6)

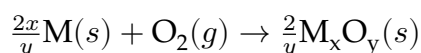
- Hardness ppm = mass of CaCO_3 equivalent per litre $\times 10^6$.
- Temporary hardness: bicarbonates; removed by boiling. Permanent: chlorides/sulphates; need chemical treatment.
- Boiler scale: CaCO_3 and CaSO_4 deposits reduce heat transfer and cause overheating.
- Corrosion: Fe is anode (oxidised); O_2/H^+ cathodic reduction drives current.
- Sacrificial anode: Zn/Mg corrodes preferentially protecting Fe.
- Stainless steel: passive Cr_2O_3 film blocks electrolyte contact.

7. Ellingham Diagrams and Free Energy in Metallurgy

An **Ellingham diagram** plots the standard Gibbs free energy of oxide formation ($\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$) against temperature for a series of metal–oxygen reactions. It is the primary tool for selecting a reducing agent in pyrometallurgy and for predicting the stability of oxides.

7.1 Construction and Interpretation

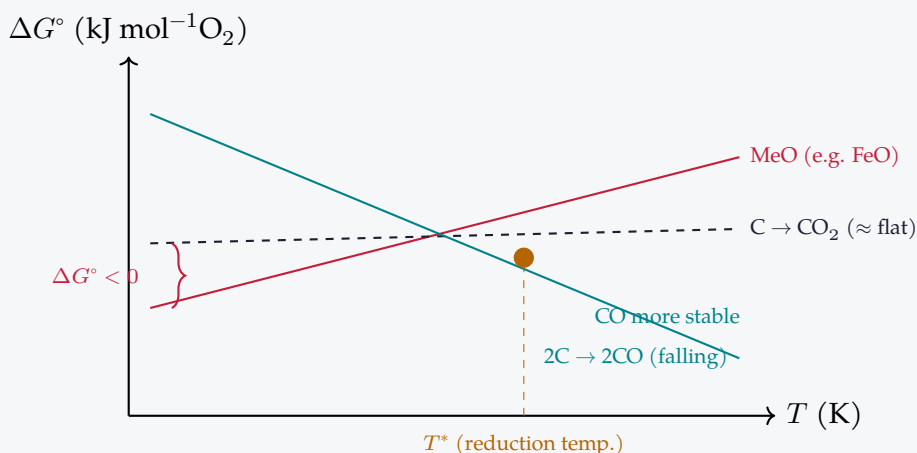
For the standard reaction written as oxidation of 1 mol O_2 :



$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ.$$

- Mostly straight lines because ΔH° and ΔS° are nearly independent of T .
- Slope = $-\Delta S^\circ_{\text{rxn}}$. Consuming one mole $O_2(g)$ makes $\Delta S^\circ < 0$ (gas disappears), so slope is *positive* (line rises with T) for most metal oxidations.
- **Exceptions:** $C + O_2 \rightarrow CO_2$: $\Delta n_{\text{gas}} = 0$, nearly horizontal. $2C + O_2 \rightarrow 2CO$: $\Delta n_{\text{gas}} = +1$, slope is *negative* (line falls with T).
- **Slope changes:** A kink (sudden change in slope) occurs at the melting or boiling point of the metal because ΔS° changes (latent heat term).

Schematic Ellingham Diagram (three key lines)



7.2 Rules for Selecting a Reducing Agent

Ellingham Selection Rule

A metal oxide MO can be reduced by a reducing agent R at temperature T if the ΔG° line of $R + O_2 \rightarrow RO_2$ lies *below* the ΔG° line of $2M + O_2 \rightarrow 2MO$ at that temperature. The overall reaction $MO + R \rightarrow M + RO$ has $\Delta G^\circ = \Delta G^\circ(\text{MO line}) - \Delta G^\circ(\text{RO line}) > 0$... wait — actually *subtract lower from upper*: net $\Delta G^\circ < 0$ only when the reducing agent's line is lower.

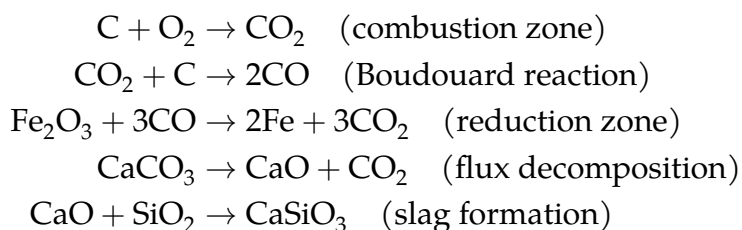
In short: **the oxide whose line is lower is more stable; its metal is harder to reduce.**

Use a reducing agent whose line is below the ore's line at the operating temperature.

7.3 Carbon as a Reducing Agent: The Blast Furnace

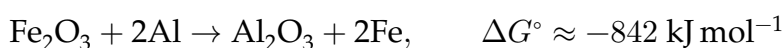
Carbon has two Ellingham lines: $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$ (flat) and $2\text{C} + \text{O}_2 \rightarrow 2\text{CO}$ (descending). Above $\sim 700^\circ\text{C}$ the CO line lies below most metal oxide lines, making carbon (via CO) an effective industrial reducing agent.

Blast furnace reactions:



7.4 Thermite Reaction: Aluminium Reducing Iron Oxide

Aluminium sits below iron oxide on the Ellingham diagram at room temperature, so:



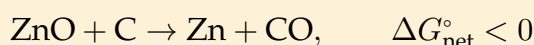
The large negative ΔG° means the reaction is strongly spontaneous and highly exothermic ($\Delta H \approx -847 \text{ kJ mol}^{-1}$), producing liquid iron at $\sim 2500 \text{ K}$. Used in rail welding and incendiary applications.

Solved Example: Reading the Ellingham Diagram

Problem: At what minimum temperature can coke (carbon) reduce ZnO?

Given: The ΔG° lines of $2\text{Zn} + \text{O}_2 \rightarrow 2\text{ZnO}$ and $2\text{C} + \text{O}_2 \rightarrow 2\text{CO}$ cross at approximately $T^* = 950^\circ\text{C}$.

Above 950°C : $\Delta G^\circ(\text{C} \rightarrow \text{CO}) < \Delta G^\circ(\text{Zn} \rightarrow \text{ZnO})$



Industrial zinc smelters operate at $\sim 1000\text{--}1100^\circ\text{C}$.

7.5 Limitations of Ellingham Diagrams

- Assumes pure solid/liquid phases (unit activity); does not account for alloy formation.
- Does not give information about reaction rate (kinetics).
- Phase changes (melting, boiling) cause kinks but data is not always available for all phases.
- Atmospheric composition (partial pressure of O_2) affects actual ΔG .

7.6 Industrial Extraction Examples Using Ellingham Diagram

Metal	Ore	Reducing agent at T	Typical temperature
Iron (Fe)	Fe_2O_3 (haematite)	C (as CO) in blast furnace	800–1600°C
Copper (Cu)	Cu_2S (chalcocite)	Roasting to Cu_2O then C or self-reduction	$\sim 1000^\circ\text{C}$
Zinc (Zn)	ZnO (from roasting ZnS)	C (as CO/C)	$\sim 1000^\circ\text{C}$
Lead (Pb)	PbO (from roasting galena)	C	$\sim 700^\circ\text{C}$
Aluminium (Al)	Al_2O_3 (bauxite)	Electrolysis (too stable for C reduction)	Hall-Héroult at 960°C

Why aluminium cannot be reduced by carbon from Ellingham: The Al_2O_3 line is always below the CO line even at 2000°C . The reaction $\text{Al}_2\text{O}_3 + \text{C} \rightarrow \text{Al} + \text{CO}$ has $\Delta G^\circ > 0$ at all practical temperatures, so electrolysis is used instead.

Exam Points (Section 7)

- Ellingham line slope = $-\Delta S^\circ$; positive (rising) for most metal oxidations; negative for $2\text{C} \rightarrow 2\text{CO}$.
- Reducing agent selection: line below = more stable oxide = better reducing agent.
- Blast furnace: C converts to CO via Boudouard reaction; CO reduces Fe_2O_3 above $\sim 700^\circ\text{C}$.
- Thermite: Al reduces Fe_2O_3 ; $\Delta G^\circ \approx -842 \text{ kJ mol}^{-1}$.
- Kink in a line: phase change (melting/boiling of metal).

Quick Revision — Module 4

Free Energy and Chemical Equilibria | BS-CH101

Key Formulas at a Glance

Formula	Meaning
$\Delta U = q + w$	First law: energy conserved.
$\Delta H = \Delta U + \Delta(PV)$	Enthalpy; at const P : $q_P = \Delta H$.
$C_P - C_V = R$	Ideal gas heat capacity relation.
$\Delta H_{\text{rxn}} = \sum \Delta H_f^\circ(\text{prod}) - \sum \Delta H_f^\circ(\text{react})$	Hess's law.
$\Delta S = \int \frac{\delta q_{\text{rev}}}{T}$	Entropy from reversible heat.
$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} \geq 0$	Second law.
$\Delta G = \Delta H - T\Delta S$	Gibbs free energy criterion.
$T^* = \Delta H / \Delta S$	Crossover temperature ($\Delta G = 0$).
$\left(\frac{\partial(\Delta G/T)}{\partial T} \right)_P = -\frac{\Delta H}{T^2}$	Gibbs–Helmholtz equation.
$\Delta G = \Delta G^\circ + RT \ln Q$	Free energy at non-standard conditions.
$\Delta G^\circ = -RT \ln K$	Standard free energy and equilibrium constant.
$\ln \frac{K_2}{K_1} = -\frac{\Delta H^\circ}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$	Van't Hoff equation.
$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$	Standard cell EMF.
$\Delta G = -nFE$; $\Delta G^\circ = -nFE^\circ$	Free energy and EMF.
$E = E^\circ - \frac{0.0591}{n} \log Q$	Nernst equation at 298 K.
$\log K = \frac{nE^\circ}{0.0591}$	K from standard cell EMF.

Key Formulas at a Glance (contd.)

$K_w = [\text{H}^+][\text{OH}^-] = 10^{-14}$	Water ionic product (298 K).
$\text{pH} = \frac{1}{2}(\text{p}K_a - \log C)$	pH of weak acid solution.
$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$	Henderson–Hasselbalch (buffer).
$K_{sp} = [\text{M}^{n+}]^m [\text{X}^{m-}]^n$	Solubility product.
$K_p = K_c (RT)^{\Delta n_g}$	Relationship between K_p and K_c .
$w_{\text{rev}} = -nRT \ln(V_2/V_1)$	Reversible isothermal expansion work.
$\Delta S_{\text{vap}} = \Delta H_{\text{vap}}/T_b$	Entropy of vaporisation.
$\frac{\Delta H_{T_2}}{\Delta H_{T_1}} = \Delta C_P (T_2 - T_1)$	Kirchhoff's equation.
$s = \sqrt{K_{sp}}$	Molar solubility of 1:1 salt.

One-Line Definitions

- **Internal energy (U):** Total kinetic and potential energy of all molecules in a system.
- **Enthalpy (H):** $H = U + PV$; heat flow at constant pressure.
- **Heat capacity (C_P):** Heat required to raise temperature by 1 K at constant pressure.
- **Hess's law:** Enthalpy change is independent of reaction pathway.
- **Entropy (S):** Measure of energy dispersal; $\Delta S_{\text{univ}} > 0$ for spontaneous processes.
- **Third law:** $S = 0$ for a perfect crystal at $T = 0$ K.
- **Gibbs free energy (G):** $G = H - TS$; $\Delta G < 0$ for spontaneous process at const T, P .
- **Crossover temperature (T^*):** Temperature at which $\Delta G = 0$; $T^* = \Delta H/\Delta S$.
- **Reaction quotient (Q):** Ratio of product to reactant activities at current conditions.
- **Equilibrium constant (K):** Q at equilibrium; $\Delta G^\circ = -RT \ln K$.
- **Van't Hoff equation:** Describes temperature dependence of K ; derived from $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$.
- **Le Chatelier's principle:** System shifts to partially counteract any imposed change.
- **Standard hydrogen electrode (SHE):** Reference electrode, $E^\circ = 0.000$ V.
- **Nernst equation:** $E = E^\circ - (RT/nF) \ln Q$; relates cell potential to concentration.
- **Faraday constant (F):** 96485 C mol^{-1} ; charge per mole of electrons.
- **Concentration cell:** Cell with identical electrodes at different concentrations; EMF from $\log(C_1/C_2)$.
- **Brønsted–Lowry acid:** Proton donor; its conjugate base forms after proton loss.
- **Buffer solution:** Resists pH change; contains weak acid + conjugate base near $\text{p}K_a$.

- **Henderson–Hasselbalch:** $\text{pH} = \text{pK}_a + \log([A^-]/[HA])$.
- **Solubility product (K_{sp}):** Ion activity product in a saturated solution.
- **Common ion effect:** Adding a common ion suppresses solubility.
- **Temporary hardness:** Due to bicarbonates; removed by boiling.
- **Permanent hardness:** Due to chlorides/sulphates; requires chemical treatment.
- **Scale:** Hard insulating deposit in boilers; reduces heat transfer.
- **Galvanisation:** Coating iron with zinc; Zn acts as sacrificial anode.
- **Ellingham diagram:** ΔG° vs T plot for metal oxide formation; used to select reducing agents.
- **Passivation:** Dense oxide film that blocks further corrosion on a metal surface.
- **BOD:** Biochemical oxygen demand; oxygen consumed by microbes to decompose organics in 5 days at 20°C .
- **COD:** Chemical oxygen demand; oxygen equivalent to oxidise all matter chemically; $\text{COD} \geq \text{BOD}$.
- **Lewis acid:** Electron-pair acceptor (e.g. BF_3 , metal ions).
- **Lewis base:** Electron-pair donor (e.g. NH_3 , H_2O).
- **Disproportionation:** Same element simultaneously oxidised and reduced (e.g. $2\text{Cu}^+ \rightarrow \text{Cu} + \text{Cu}^{2+}$).
- K_p and K_c : Related by $K_p = K_c(RT)^{\Delta n_g}$; equal when $\Delta n_g = 0$.
- **Kirchhoff's equation:** ΔH varies with temperature as $d(\Delta H)/dT = \Delta C_P$.
- **Trouton's rule:** $\Delta S_{\text{vap}} \approx 85 \text{ J mol}^{-1} \text{ K}^{-1}$ for non-associated liquids at their normal boiling point.

Frequently Asked / Viva Questions

Q1. State the first law of thermodynamics.

$\Delta U = q + w$; energy is conserved; heat and work are equivalent forms of energy transfer.

Q2. What is enthalpy and why is it useful?

$H = U + PV$; at constant pressure, $\Delta H = q_P$ — directly measurable as heat flow in open experiments.

Q3. State Hess's law with one example.

Enthalpy change is independent of pathway. Example: $\Delta H_{\text{combustion}}$ of C to CO derived from ΔH of $\text{C} \rightarrow \text{CO}_2$ and $\text{CO} \rightarrow \text{CO}_2$.

Q4. State the second law of thermodynamics.

For any spontaneous process, $\Delta S_{\text{universe}} > 0$; for reversible processes = 0.

Frequently Asked / Viva Questions (contd.)

Q5. What is Gibbs free energy and its spontaneity criterion?

$G = H - TS$; at constant T and P : $\Delta G < 0$ spontaneous, $\Delta G = 0$ equilibrium, $\Delta G > 0$ non-spontaneous.

Q6. List the four cases of ΔH and ΔS signs.

$(-, +)$: always spontaneous. $(+, -)$: never. $(-, -)$: spontaneous at low T . $(+, +)$: spontaneous at high T .

Q7. Derive $\Delta G^\circ = -RT \ln K$.

At equilibrium, $\Delta G = 0$ and $Q = K$. Substituting in $\Delta G = \Delta G^\circ + RT \ln Q$ gives $0 = \Delta G^\circ + RT \ln K$, so $\Delta G^\circ = -RT \ln K$.

Q8. State the van't Hoff equation.

$d \ln K / dT = \Delta H^\circ / (RT^2)$; exothermic reactions have K decreasing with T .

Q9. What is the Nernst equation?

$E = E^\circ - (RT/nF) \ln Q = E^\circ - (0.0591/n) \log Q$ at 298 K; it accounts for concentration effects on cell EMF.

Q10. How is K found from E° ?

$\log K = nE^\circ / 0.0591$ at 298 K; derived from $\Delta G^\circ = -nFE^\circ = -RT \ln K$.

Q11. What is a concentration cell?

Cell with same electrode in solutions of different concentrations; $E = (0.0591/n) \log(C_{\text{high}}/C_{\text{low}})$.

Q12. What is the Henderson–Hasselbalch equation?

$\text{pH} = \text{pK}_a + \log([A^-]/[HA])$; gives pH of a buffer; maximum capacity at $\text{pH} = \text{pK}_a$.

Q13. Derive the pH of a weak acid solution.

$K_a = x^2 / (C - x) \approx x^2 / C$; $x = \sqrt{K_a C}$; $\text{pH} = \frac{1}{2}(\text{pK}_a - \log C)$.

Q14. What is K_{sp} and when does precipitation occur?

Ion activity product at saturation; precipitation occurs when ionic product $> K_{sp}$.

Q15. Explain the common ion effect on solubility.

Adding a common ion shifts the dissolution equilibrium backwards, reducing solubility (BaSO_4 in Na_2SO_4 solution).

Q16. What is temporary and permanent hardness?

Temporary: bicarbonates, removed by boiling. Permanent: chlorides/sulphates, need lime-soda or ion exchange.

Frequently Asked / Viva Questions (contd.)

Q17. Explain the mechanism of iron corrosion.

Iron acts as anode: $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$; cathode reduces O_2 : $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$; products react to form $\text{Fe}(\text{OH})_3$ (rust).

Q18. What is galvanisation and how does it protect iron?

Coating iron with zinc; even if coating is scratched, Zn (more active, lower E°) acts as sacrificial anode and corrodes preferentially.

Q19. What does an Ellingham diagram show?

ΔG° vs T for metal oxide formation; lower line = more thermodynamically stable oxide.

Q20. Why does the $2\text{C} \rightarrow 2\text{CO}$ line fall in the Ellingham diagram?

The reaction $2\text{C}(s) + \text{O}_2(g) \rightarrow 2\text{CO}(g)$ produces more moles of gas than it consumes, so $\Delta S > 0$ and the line has negative slope: ΔG° decreases as T rises.

Q21. What is the Boudouard reaction?

$\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$; above $\sim 700^\circ\text{C}$, CO becomes the dominant reducing gas in the blast furnace.

Q22. Explain the thermite reaction.

$\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Fe}$; Al forms a more stable oxide than Fe, making $\Delta G^\circ \approx -842 \text{ kJ mol}^{-1}$; extremely exothermic.

Q23. What is BOD and COD?

BOD = biochemical oxygen demand (O_2 consumed by microbes in 5 days at 20°C); COD = chemical oxygen demand (total O_2 equivalent for all oxidisable matter); $\text{COD} \geq \text{BOD}$ always; high values indicate heavy organic pollution.

Q24. Why can carbon not reduce aluminium oxide?

The Al_2O_3 Ellingham line is always below the $\text{C} \rightarrow \text{CO}$ line at all practical temperatures, so $\Delta G^\circ > 0$ for the reduction. Electrolysis (Hall-Hérault) is used instead.

Q25. What is passivation?

Formation of a dense, adherent oxide film that prevents further corrosion. Stainless steel ($> 11\% \text{ Cr}$) forms Cr_2O_3 ; Al forms Al_2O_3 .

Q26. Relate K_p and K_c .

$K_p = K_c (RT)^{\Delta n_g}$; they are equal when $\Delta n_g = 0$ (no change in moles of gas).

Q27. State Trouton's rule.

Most non-associated liquids have $\Delta S_{\text{vap}} \approx 85 \text{ J mol}^{-1} \text{ K}^{-1}$ at their boiling point; water deviates ($109 \text{ J mol}^{-1} \text{ K}^{-1}$) due to H-bonding.

Frequently Asked / Viva Questions (contd.)

Q28. What is a Lewis acid? Give an example.

Electron-pair acceptor; e.g. BF_3 accepts the lone pair of NH_3 to form $\text{F}_3\text{B}:\text{NH}_3$.

Q29. What is disproportionation?

A reaction where the same element is simultaneously oxidised and reduced; e.g. $2\text{Cu}^+(\text{aq}) \rightarrow \text{Cu}(\text{s}) + \text{Cu}^{2+}(\text{aq})$; spontaneous because $E^\circ = 0.52 - 0.16 = +0.36 \text{ V}$.

Q30. Write the $\text{H}_2\text{-O}_2$ fuel cell reactions.

Anode: $2\text{H}_2 + 4\text{OH}^- \rightarrow 4\text{H}_2\text{O} + 4\text{e}^-$. Cathode: $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$. Overall: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, $E^\circ = 1.23 \text{ V}$.

Summary Reference Table — Module 4

Topic	Key concept	Key formula	Exam output
First law	Energy conserved	$\Delta U = q + w$	Hess's law cycle
Second law	Entropy increases	$\Delta S_{\text{univ}} \geq 0$	Entropy calculation
Gibbs free energy	Spontaneity at const T , P	$\Delta G = \Delta H - T\Delta S$	Four-case table + T^*
Equilibrium	ΔG minimum	$\Delta G^\circ = -RT \ln K$	K from ΔG°
Van't Hoff	K vs temperature	$\ln(K_2/K_1) = \dots$	Temperature extrapolation
Electrochemistry	$\Delta G = -nFE$	Nernst equation	Cell EMF numerical
Acid-base	Buffer / weak acid	Henderson-Hasselbalch	pH calculation
K_{sp} / solubility	Precipitation	IP vs K_{sp}	Solubility with common ion
Water chemistry	Hardness, boiler troubles	ppm as CaCO_3	Scale / sludge / priming
Corrosion	Electrochemical oxidation	Anodic / cathodic reactions	Prevention methods table
Ellingham diagram	Oxide stability vs T	Lower line = more stable	Reducing agent selection